MSc PROJECT IMPLEMENTATION & STATUS GUIDE

AI-Driven Weather Forecasting and Crop Recommendation System for Anuradhapura District

Document purpose	Readable project handover and gap assessment
Based on	Chapters 1-4, working PRD/SRS baseline, evidence notes and latest reported Codex QA
Status date	17 October 2026
Audience	MSc supervisor, examiner, developer and project owner

The project in one minute

This is a web-based agricultural decision-support prototype for the whole Anuradhapura District. It predicts rainfall, temperature and relative humidity for the next seven days using a project-developed LSTM model. The system shows forecasts in an Angular web application and attempts to help users understand which of six crops may be appropriate, while clearly identifying where scientific evidence is insufficient.

Current overall position

Area	Evidence-based position
Weather forecasting	WORKING - trained LSTM, live weather pipeline and stored forecasts reported tested
Model performance	EVALUATED - LSTM outperformed a persistence baseline on all three variables
Crop recommendation	PARTIAL - full six-crop suitability scores/categories/ranking not scientifically supported
Users and roles	Public, Registered User and Administrator tested; Officer live validation blocked
Readiness	Functional prototype, not yet a fully validated crop recommendation product

Important reading rule: "Implemented" means code or a workflow exists; "Tested" means a reported QA result; "Scientifically validated" requires appropriate agricultural evidence and review. These are different levels of confidence.

1. What we originally proposed

The Chapters 1-4 dissertation document is the principal design narrative. The separate PRD/SRS source-of-truth file is a working draft, not independently verified as supervisor-approved. The requirements below reflect the original design, with later seasonal-planning additions identified separately.

Requirement	What success would look like
Area and crops	Whole Anuradhapura District; Paddy, Maize, Green Gram, Cowpea, Groundnut and Chilli
Weather AI	Our own LSTM, 30 preceding days of weather input, seven-day rainfall/temperature/humidity output
Crop suitability	Factor-level assessment of rainfall, temperature, humidity and optional soil; weighted overall score, category, crop ranking and explanation
Users	Public access; Registered User; Agricultural Officer for review/validation; Administrator for management
Data and history	Persist forecasts, crop configurations, recommendations and applicable validation/history records
Quality and research	Security, explainability, traceability, testing and chronological MAE/RMSE evaluation

Later requested seasonal-planning functions

Two extra user workflows were developed/reported after the original Chapter 3/4 baseline: date range to crop candidates and crop to candidate cultivation periods. They use historical climate and official crop-calendar evidence. They are not seven-day LSTM predictions, and formal supervisor approval as original scope is not independently documented.

What is deliberately outside the core model

The project does not claim that its seven-day LSTM predicts months or seasons ahead. It does not provide field-level precision, yield prediction, irrigation optimization or an independently trained long-term weather model. Soil data is optional, not a condition for using the system.

2. How the system is connected

Think of the project as five connected parts. The web page asks for information; the API controls the work; the forecasting service runs the AI model; SQL Server stores important records; and Open-Meteo supplies recent weather observations.

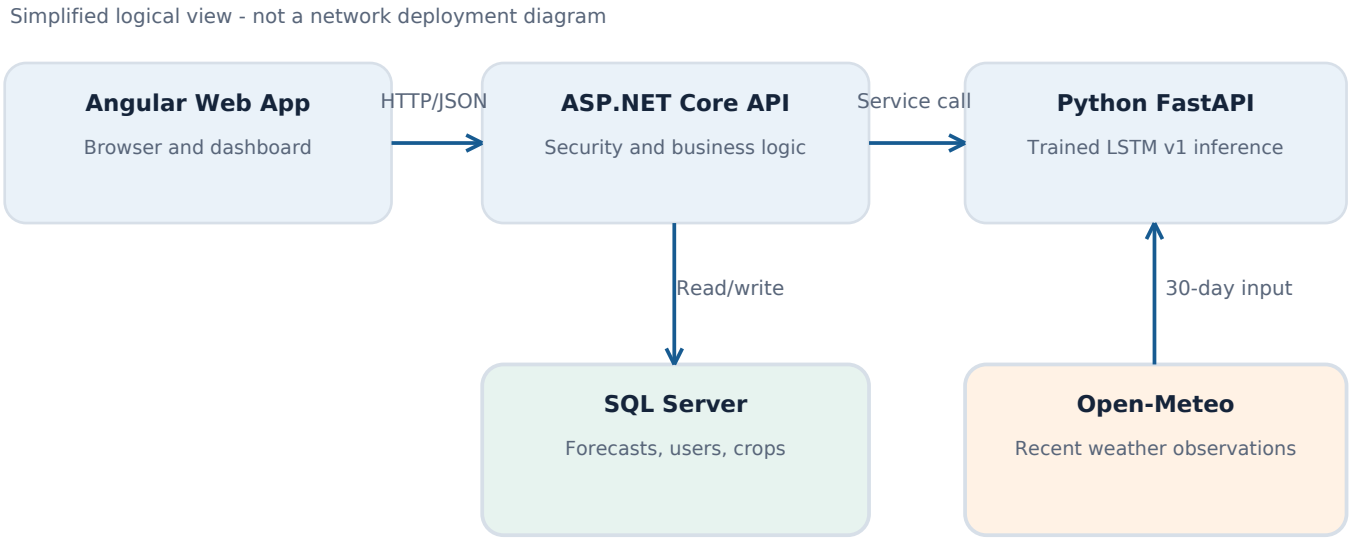


Figure 1. Simplified integration overview. The API mediates application access; the browser does not directly connect to SQL Server or the LSTM model.

Component	What it does	Technology
Frontend	Pages, forms, dashboards, result explanations	Angular
Backend	HTTP APIs, authentication, permissions, business logic	ASP.NET Core
AI service	Loads trained model and generates seven-day predictions	Python FastAPI / TensorFlow LSTM
Database	Stores user roles, crop information, forecast runs and related records	SQL Server
Weather source	Provides recent observed weather for operational inference	Open-Meteo

One rule to remember

Angular talks to ASP.NET Core. The ASP.NET Core backend communicates with the Python model service and SQL Server. This separation is what allows the same forecast data to be displayed, saved and reused by recommendation workflows.

3. What we built - step by step

A. Train and evaluate the AI model

1. Gather historical ERA5 weather data for 2010-2025. 2. Prepare daily rainfall, mean temperature and relative humidity. 3. Convert data into 30-day input sequences. 4. Train an LSTM to predict the next seven days for all three variables. 5. Save the trained v1.keras model, scaler and metadata. 6. Evaluate using chronological held-out data, not a random future/past mixture.

B. Generate a new operational forecast

1. An authorized administrator triggers operational forecast generation. 2. Open-Meteo provides recent weather observations. 3. The backend prepares the last 30 days for the model. 4. The Python FastAPI service loads LSTM v1 and returns seven dated predictions. 5. The backend saves a ForecastRun and forecast records in SQL Server. 6. The latest-forecast API returns stored predictions to the Angular dashboard.

C. Show crop information

1. A public user opens the dashboard or crop assessment page. 2. Angular sends a request to the ASP.NET Core recommendation API. 3. The backend retrieves the forecast and supported crop data. 4. It evaluates only those crop factors supported by evidence. 5. Angular displays the result and explains missing factors. It currently uses PARTIALLY_ASSESSED rather than inventing a complete crop ranking.

D. Seasonal planning - separate evidence path

The user can select a date range or a crop. The system uses historical climate patterns and Sri Lankan Department of Agriculture crop calendars to present conditional cultivation-period candidates. This is historical/calendar planning, not a 3-, 6- or 12-month LSTM forecast.

Example: verified operational run

Detail	Reported evidence
Forecast dates	17-23 October 2026 (seven days)
Path	Open-Meteo -> LSTM v1 -> SQL Server -> Angular -> crop evaluation
Result	Stored forecast and public crop evaluation reported PASS; scientific suitability remained PARTIAL

4. Model evaluation - what the numbers say

The trained model was tested on 725 chronological sequences from 1 January 2024 to 31 December 2025. A simple persistence baseline was used for comparison. MAE and RMSE are error measures: lower values are better. These results were supplied by the latest Codex evaluation report.

Variable	LSTM MAE	Baseline MAE	LSTM RMSE	Baseline RMSE
Rainfall (mm)	4.9042	5.4930	10.5467	13.9157
Temperature (C)	0.7071	0.7930	0.8912	1.0405
Humidity (%)	3.8786	4.4646	4.8813	5.8587

Variable	MAE reduction	RMSE reduction	Interpretation
Rainfall	10.72%	24.21%	Improves on baseline; remains highest-error variable
Temperature	10.83%	14.34%	Improves on baseline; smallest absolute error
Humidity	13.13%	16.68%	Improves on baseline for both metrics

What we can conclude

The project-developed LSTM outperformed the selected persistence baseline on all three measured weather variables. This supports the usefulness of the developed model relative to that baseline and on that test set. It does not prove perfect forecasts, universal superiority over all models or reliable agronomic suitability classification. The evidence does not currently require retraining.

Why forecasting success does not finish crop recommendations

The model predicts weather. A recommendation engine must then interpret those predictions using crop-specific, time-scale-compatible agricultural requirements. For example, annual rainfall requirements cannot simply be compared with a seven-day rainfall total. This is why model accuracy and recommendation validity must be evaluated separately.

5. Implemented requirements vs remaining gaps

Scope item	Current reported result	Status
LSTM training and seven-day inference	Model v1 saved, loaded and executed	PASS
Historical evaluation and baseline	MAE/RMSE and comparison completed	PASS
Operational pipeline	Open-Meteo -> Python -> SQL -> API -> Angular	PASS
Public dashboard and seasonal pages	Visible workflows and input validation tested	PASS
Registration and role protection	Registered User and Admin tested; no self-escalation	PASS
Six-crop catalogue	All six crops represented	PASS
Crop suitability scores/categories/rank	Only supported temperature indicator; no full ranking	PARTIAL
Agricultural Officer live validation	Existing Officer account, but no authorized test access	BLOCKED
Recommendation history	Disabled in current local configuration; enabled workflow not tested	NOT VERIFIED
Climate-risk indicators	Conditional feature; enabled-path acceptance not evidenced	NOT VERIFIED
Multilingual support	Not evidenced; scope status needs confirmation	NOT VERIFIED
Security dependency warning	Transitive Pkcs 9.0.11 advisory; fix pending approval	OPEN

QA snapshot from the most recent report: 33 cases - 31 PASS, 1 PARTIAL, 1 BLOCKED, 0 FAIL. Regression suites: backend 289/289, Python 22/22, Angular 39/39. These figures describe reported tests, not an independent repository audit.

6. Why the crop recommendation is still PARTIAL

The original design asks for an overall suitability score and categories such as Highly Suitable, Suitable, Moderately Suitable and Unsuitable, based on rainfall, temperature, humidity and optional soil compatibility. This outcome has not been scientifically demonstrated.

Factor	Current evidence	Correct handling
Temperature	General ecological temperature compatibility available	Show a labelled partial indicator only
Rainfall	No validated local seven-day crop-specific thresholds	Do not compute an unsupported suitability score
Humidity	No validated local seven-day crop-specific thresholds	Do not compute an unsupported suitability score
Soil	Optional; site-specific evidence not supplied	Mark unknown; do not assume good or bad soil
Overall ranking	Comparable complete factor evidence unavailable	Do not assign an overall category or rank

What the system should tell a user

"This crop has been partially assessed using available temperature evidence. A complete suitability conclusion cannot be made because locally validated short-term rainfall and humidity criteria and/or other required evidence are unavailable."

What would close the research gap

A defensible scoring specification requires locally relevant crop criteria, correct measurement units and time periods, documented factor weights and missing-data policy, and qualified agricultural expert review. If that evidence cannot be obtained within the MSc timeline, the correct academic outcome is to report the limitation, not fabricate thresholds.

Seasonal candidates are not a substitute

Maha/Yala calendar-supported candidate periods can be useful decision support. However, a calendar-supported cultivation window is not equivalent to a validated numerical seven-day crop suitability ranking.

7. Users, database and security

What each role is for

Role	Intended capability	Reported QA
Public	View weather and use public crop-related pages	PASS
Registered User	Register, log in, access own authorized features	PASS for registration/auth; history disabled
Agricultural Officer	Review/validate recommendations; read-only configuration	Access controls audited; live validation BLOCKED
Administrator	Manage users/configuration and generate forecasts	PASS for tested admin paths

What the database holds

The Chapter 4 design names UserRole, User, Crop, CropEnvironmentalRequirement, SoilCompatibility, SuitabilityConfiguration, ForecastRecord, Recommendation, RecommendationCrop and RecommendationValidation. The running system also reports persisted operational ForecastRun records. These are design entities and reported implementation concepts; exact live SQL schema and record counts require a separate database inspection.

Connection and credentials

The backend was reported to connect successfully to the existing SQL Server; health and latest-forecast endpoints returned successfully. The database password was moved to local ASP.NET Core user-secrets. No IP addresses, usernames, passwords or full connection strings are included in this guide.

Outstanding security maintenance

A transitive Microsoft.Data.SqlClient -> System.Security.Cryptography.Pkcs 9.0.11 vulnerability warning was reported. A narrow, compatible patched dependency update was proposed, but it still requires authorization and verification. The protected API project file must remain untouched without explicit permission.

8. Exactly what remains to finish

Priority	Action	Completion evidence
1 - Technical	Authorized Agricultural Officer validation test with safe test recommendation; use approved local test configuration only	Officer action, validation status/comment, persistence and role restrictions demonstrated
2 - Scientific	Review QA-001 with supervisor/agricultural expert; either validate a defensible scoring method or formally retain the limitation	Signed/reviewed methodology decision and transparent partial-output evidence
3 - Security	Approve and apply narrow compatible Pkcs dependency patch without touching protected Api.csproj	Clean vulnerability audit and passing regression tests
4 - Scope audit	Confirm history, climate-risk and multilingual expectations against final approved scope; test only requirements actually in scope	Requirements-to-QA traceability with PASS / CONDITIONAL / NOT VERIFIED
5 - Academic	Finish Chapters 5-8 using actual implementation, model metrics, comparison, QA and limitations	Development, Testing, Evaluation and Conclusion chapters reviewed
6 - Closeout	Perform final regression, back up necessary evidence and remove only verified disposable temporary files	Final QA summary and safe cleanup log

Do not spend time repeating these completed tasks

Do not retrain the LSTM without evidence of a defect; do not repeat the completed baseline comparison; do not repeat the same unsuccessful crop-threshold search; do not add unapproved features; do not repeatedly toggle working development feature flags; do not alter production settings or delete project evidence.

Project-owner decisions still needed

A. Provide authorized Officer test access and a safe pending recommendation. B. Confirm whether Officer validation may be enabled in local test configuration. C. Approve the narrow security dependency update. D. Obtain a supervisor decision on the crop-suitability scientific limitation and any later-added seasonal-planning scope.

9. Conclusion and evidence references

Present conclusion

The project has delivered a technically functioning, integrated web-based weather forecasting prototype for Anuradhapura District. Its project-developed LSTM produces seven-day rainfall, temperature and humidity predictions, and its measured errors were lower than a persistence baseline on the reported chronological test data. Forecast generation, SQL persistence, API access and Angular display have been exercised end to end.

However, the original complete crop suitability objective is only partially achieved: the system does not yet have scientifically validated, locally applicable short-horizon rainfall and humidity scoring criteria for a defensible six-crop overall ranking. Live Agricultural Officer validation, selected conditional functions and security maintenance also remain open. Therefore the most accurate project status is functional prototype with incomplete agronomic validation, not a fully validated agricultural recommendation product.

Source documents used

Source	How used
Chapters_1_to_4_Piyumi_Paramee_Darshika.docx	Original aim, geographic/crop scope, user requirements, architecture, database and crop suitability design
PROJECT_SOURCE_OF_TRUTH_v1.md	Working PRD/SRS and traceability baseline; explicitly draft, not verified supervisor-approved
STAGE2_STAGE3_EVIDENCE_AND_SCORING_SPEC.md	Evidence boundaries for temperature, rainfall, humidity, soil and suitability scoring
DOA_SIX_CROP_CALENDAR_EXTRACTION_AUDIT.md	Crop calendar extraction and conditional seasonal-planning provenance
Recent Codex reports provided in chat	Current implementation, end-to-end tests, evaluation metrics, SQL configuration and QA snapshot
Earlier QA_TEST_CASES.md and PROPOSED_VS_ACTUAL_WORKFLOW_AUDIT.md	Historical QA/design gap context; older statuses superseded by later reports

Evidence and approval caveat

This guide combines accessible project documents with user-supplied Codex reports dated through 17 October 2026. The current source repository, live database and latest local PDF were not independently inspected in preparing this guide. Any item marked PASS is therefore a reported result. Final dissertation claims should be backed by retained logs, test evidence, screenshots, source files, model artefacts and supervisor review.

Use of this document: Orientation, supervisor discussion, project handover and final work planning. It does not replace the signed-off proposal, authoritative requirements, validated agronomic evidence or final dissertation.